From Kaprekar Dynamics to Logit Geometry:

Exact Gap-Spectral Reduction Theorems and a Benchmark-Ready Representation for LLM Reliability

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Abstract

Kaprekar's constant 6174 is a special case of a richer finite dynamical system. This paper isolates a gap-spectral formulation of Kaprekar's routine and then transfers the same geometry to large-language-model (LLM) logit spectra. For the base-b, n-digit Kaprekar map $K_{b,n}$, we prove an exact factorization through the ordered outer-digit gaps, obtaining a reduced state space of dimension $\lfloor n/2 \rfloor$. This yields an explicit description of the one-step image and the exact image-cardinality formula

$$|\operatorname{Im}(K_{b,n})| = \binom{b + \lfloor n/2 \rfloor - 1}{\lfloor n/2 \rfloor}.$$

Thus, for fixed base, the effective one-step state space is polynomial in n rather than exponential in the raw digit count. Motivated by current LLM uncertainty work, we introduce the *Kaprekar gap simplex* for sorted top-k logits and prove that it parametrizes the affine quotient of sorted logit spectra. The resulting coordinates give a mathematically principled representation for uncertainty heads, routing, and relaxed speculative verification. We also prove a no-free-lunch lemma showing that no score depending only on current logits can be universally sufficient for correctness, which motivates hybridization with internal-state probes and conformal calibration. The paper is theorem-first and benchmark-ready: all proved claims are self-contained, and all deployment claims are stated as falsifiable hypotheses rather than as unsupported state-of-the-art assertions.

Keywords. Kaprekar routine; discrete dynamical systems; fixed points; uncertainty quantification; large language models; speculative decoding; conformal calibration.

1 Introduction

Kaprekar's 1949 and 1955 papers initiated a family of digit-rearrangement maps whose most famous decimal fixed point is 6174 [1, 2]. Subsequent work classified the classical decimal constants 495 and 6174, studied loops and convergence, generalized to other bases and digit lengths, and recently revisited the routine through symmetry, entropy, and finite-dynamical-systems viewpoints [3, 4, 5, 6, 8, 10, 11, 13, 14, 15, 16, 17]. The mature lesson of this literature is that 6174 is not the final object: beyond three and four decimal digits one typically encounters multiple fixed points or genuine cycles rather than a unique universal constant.

In parallel, the current LLM-uncertainty literature has moved away from naive confidence surrogates toward richer signals: semantic entropy and its cheaper probes [19, 20], logit-based token uncertainty [22], query-level uncertainty before answering [23], hidden-state and attention-based correctness probes [18, 27, 28, 30, 31], and selective-prediction or conformal wrappers that explicitly target trust calibration under deployment constraints [21, 24, 29, 37]. Recent speculative-decoding work has likewise shifted attention from draft/target alignment alone to learned judges, flat-head regimes, and direct acceptance-rate objectives [25, 36, 34].

This paper connects those two threads. The mathematical point is simple: Kaprekar's routine is naturally expressed in *ordered gap coordinates*. The machine-learning point is equally simple: sorted top-k logits admit the same kind of gap geometry. Our main contributions are:

1. We isolate an exact arbitrary-base factorization of the Kaprekar map through ordered outer-digit gaps. Related difference-coordinate ideas appear in earlier work, especially in Walden, Thakur, and Nuez [8, 13, 14]; our goal is to make the quotient structure explicit and to use it systematically.

2. We prove an exact image-cardinality theorem for one-step Kaprekar outputs, reducing the raw state space from b^n to $\binom{b+\lfloor n/2\rfloor-1}{\lfloor n/2\rfloor}$.

3. We transfer the same gap geometry to sorted top-k logits, defining the *Kaprekar gap simplex* and proving that it parametrizes the affine quotient of sorted logit spectra.

4. We derive precise, falsifiable consequences for LLM reliability: a hybrid uncertainty architecture, a no-free-lunch lemma explaining why logits alone cannot be universally sufficient, and a relaxed speculative-verification rule with an explicit probability-ratio bound.

The result is not a new "better constant." It is a better equation and a better coordinate system.

2 Historical and technical background

2.1 Kaprekar literature, 1949–2026

Kaprekar introduced the digit-rearrangement game in 1949 and highlighted the number 6174 in 1955 [1, 2]. Jordan's 1964 note studied self-producing digit sequences [3]. Hasse and Prichett classified the four-digit constants in 1978, and Prichett, Ludington, and Lapenta proved in 1981 that the only nonzero *decimal* Kaprekar constants are 495 and 6174 [4, 5]. Young's 1993 work and Deutsch–Goldman's 2004 exposition explored variants and pedagogy [6, 7]. Walden generalized the search across bases in 2005 [8]. More recent work has treated base dependence, symmetry, 2-adic and odd/even-base structure, and attractor-basin entropy [9, 10, 11, 12, 13, 14, 15, 16, 17].

2.2 LLM uncertainty and reliability, 2023–2026

A now-standard tension in LLM reliability is that cheap output-based signals are often informative yet brittle, whereas internal-state probes can be stronger but harder to calibrate and transfer. Hidden-state truthfulness probing appears already in Azaria and Mitchell [18]. Semantic entropy and semantic-entropy probes detect confabulations by reasoning in meaning-space rather than token-space [19, 20]. Benchmarking efforts such as LM-Polygraph have made cross-method comparisons more systematic [21]. Recent work studies logit-based uncertainty [22], query-level uncertainty before generation [23], reasoning-model calibration [26], selected-token internal-state aggregation [27], model-internals audits [28], attention-entropy predictors [30], local-information-score methods [31], and deployment-facing selective prediction [29]. Conformal methods provide finite-sample guarantees when their assumptions hold [24, 37]. On the acceleration side, judge-based and acceptance-aware speculative decoding has become a major current direction [25, 36, 34, 32].

3 Kaprekar factorization through ordered outer-digit gaps

Fix a base $b \geq 2$ and a digit count $n \geq 1$. Every $x \in \{0, \dots, b^n - 1\}$ is represented with leading zeros as an n-tuple of digits. Let

$$a_1 \geq a_2 \geq \cdots \geq a_n$$

be the digits of x sorted in descending order, and let

$$m = \lfloor n/2 \rfloor.$$

Definition 3.1 (Kaprekar map)**.** The base-b, n-digit Kaprekar map is

$$K_{b,n}(x) = \sum_{i=1}^{n} a_i b^{n-i} - \sum_{i=1}^{n} a_i b^{i-1}.$$

Definition 3.2 (Kaprekar spectrum)**.** The *Kaprekar spectrum* of x is the weakly decreasing m-tuple

$$\Sigma(x) = (\Delta_1, \dots, \Delta_m), \qquad \Delta_j := a_j - a_{n+1-j}.$$

Thus Δ_j is the jth outer-pair gap among the sorted digits. We also define

$$\mathcal{S}_{b,n} := \{(\Delta_1, \dots, \Delta_m) \in \mathbb{Z}^m : b - 1 \geq \Delta_1 \geq \cdots \geq \Delta_m \geq 0\}.$$

Definition 3.3 (Gap-spectral linear form)**.** For $\Delta = (\Delta_1, \dots, \Delta_m) \in \mathcal{S}_{b,n}$ define

$$\Lambda_{b,n}(\Delta) := \sum_{j=1}^{m} \left(b^{n-j} - b^{j-1}\right) \Delta_j.$$

Theorem 3.4 (Exact factorization)**.** *For every x,*

$$K_{b,n}(x) = \Lambda_{b,n}(\Sigma(x)).$$

Equivalently,

$$K_{b,n} = \Lambda_{b,n} \circ \Sigma.$$

Proof. By definition,

$$K_{b,n}(x) = \sum_{i=1}^{n} a_i \left(b^{n-i} - b^{i-1}\right).$$

Pair the ith term with the $(n+1-i)$th term:

$$a_i \left(b^{n-i} - b^{i-1}\right) + a_{n+1-i} \left(b^{i-1} - b^{n-i}\right) = (a_i - a_{n+1-i}) \left(b^{n-i} - b^{i-1}\right).$$

For $1 \leq i \leq m$ this equals

$$\Delta_i \left(b^{n-i} - b^{i-1}\right).$$

If n is odd, the middle term has coefficient $b^m - b^m = 0$ and vanishes. Summing the pairs gives the formula. □

Corollary 3.5 (Reduced spectral dynamics)**.** *Define*

$$R_{b,n} := \Sigma \circ \Lambda_{b,n}.$$

Then

$$\Sigma \circ K_{b,n} = R_{b,n} \circ \Sigma.$$

Therefore fixed points and cycles of $K_{b,n}$ correspond to fixed points and cycles of the reduced map $R_{b,n}$ on $\mathcal{S}_{b,n}$.

Proof. This is an immediate composition identity:

$$\Sigma \circ K_{b,n} = \Sigma \circ \Lambda_{b,n} \circ \Sigma = R_{b,n} \circ \Sigma.$$

$\square$

Corollary 3.6 (Exact image description)**.**

$$\mathrm{Im}(K_{b,n}) = \{\Lambda_{b,n}(\Delta) : \Delta \in \mathcal{S}_{b,n}\}.$$

Proof. The inclusion $\subseteq$ follows from Theorem 3.4. For the reverse inclusion, let $\Delta \in \mathcal{S}_{b,n}$. If $n = 2m$, choose sorted digits

$$a_1 = \Delta_1, \ldots, a_m = \Delta_m, \qquad a_{m+1} = \cdots = a_{2m} = 0.$$

If $n = 2m + 1$, insert one additional middle digit $a_{m+1} = 0$. Because $\Delta_1 \geq \cdots \geq \Delta_m \geq 0$, these are valid sorted digits, and their Kaprekar spectrum is exactly Δ. Theorem 3.4 then gives $K_{b,n}(x) = \Lambda_{b,n}(\Delta)$. $\square$

Theorem 3.7 (Exact one-step image cardinality)**.** *For every base $b \geq 2$ and digit count $n \geq 1$,*

$$|\,\mathrm{Im}(K_{b,n})| = \binom{b + \lfloor n/2 \rfloor - 1}{\lfloor n/2 \rfloor}.$$

Proof. We first count $\mathcal{S}_{b,n}$. Let $m = \lfloor n/2 \rfloor$ and define

$$y_0 = b - 1 - \Delta_1, \qquad y_1 = \Delta_1 - \Delta_2, \quad \ldots, \quad y_{m-1} = \Delta_{m-1} - \Delta_m, \qquad y_m = \Delta_m.$$

Then each $y_i \geq 0$ and

$$y_0 + y_1 + \cdots + y_m = b - 1.$$

This gives a bijection between $\mathcal{S}_{b,n}$ and weak compositions of $b - 1$ into $m + 1$ parts, so

$$|\mathcal{S}_{b,n}| = \binom{(b-1) + (m+1) - 1}{m} = \binom{b + m - 1}{m}.$$

It remains to show that $\Lambda_{b,n}$ is injective on $\mathcal{S}_{b,n}$. Write

$$c_j := b^{n-j} - b^{j-1}.$$

Suppose $\Delta, \Delta' \in \mathcal{S}_{b,n}$ are distinct, and let j be the smallest index with $\Delta_j \neq \Delta'_j$. Since both tuples are weakly decreasing, after swapping the names if necessary we may assume $\Delta_j > \Delta'_j$. Then

$$\Lambda_{b,n}(\Delta) - \Lambda_{b,n}(\Delta') = \sum_{k=j}^{m} c_k(\Delta_k - \Delta'_k) \geq c_j - (b-1) \sum_{k=j+1}^{m} c_k.$$

Thus it is enough to prove

$$c_j > (b-1) \sum_{k=j+1}^{m} c_k.$$

If $n = 2m$, then

$$(b-1) \sum_{k=j+1}^{m} c_k = b^{2m-j} - 2b^m + b^j,$$

so

$$c_j - (b-1)\sum_{k=j+1}^{m} c_k = \left(b^{2m-j} - b^{j-1}\right) - \left(b^{2m-j} - 2b^m + b^j\right) = 2b^m - (b+1)b^{j-1} > 0.$$

If $n = 2m+1$, then

$$(b-1)\sum_{k=j+1}^{m} c_k = b^{2m+1-j} - (b+1)b^m + b^j,$$

so

$$c_j - (b-1)\sum_{k=j+1}^{m} c_k = \left(b^{2m+1-j} - b^{j-1}\right) - \left(b^{2m+1-j} - (b+1)b^m + b^j\right) = (b+1)\left(b^m - b^{j-1}\right) > 0.$$

Hence $\Lambda_{b,n}$ is injective, and

$$|\operatorname{Im}(K_{b,n})| = |\mathcal{S}_{b,n}| = \binom{b+m-1}{m}.$$

□

Remark 3.8 (What the theorem means)**.** For fixed base, the one-step image is polynomially sized in n. In decimal,

$$|\operatorname{Im}(K_{10,4})| = 55, \quad |\operatorname{Im}(K_{10,6})| = 220, \quad |\operatorname{Im}(K_{10,8})| = 715, \quad |\operatorname{Im}(K_{10,10})| = 2002.$$

So a ten-digit Kaprekar study begins not from 10^{10} one-step states but from exactly 2002 reduced states.

Example 3.9 (Decimal fixed spectra)**.** A reduced search over $\mathcal{S}_{10,n}$ immediately reproduces familiar facts and clarifies why there is no universal replacement for 6174. Table 1 lists the nonzero decimal fixed spectra for small n.

Table 1: Nonzero decimal fixed spectra recovered from the reduced map $R_{10,n}$.

n	Nonzero fixed spectra Δ	Fixed point(s)
3	(5)	495
4	$(6,2)$	6174
5	none	none
6	$(6,3,2)$, $(5,5,0)$	631764, 549945
7	none	none
8	$(9,7,5,1)$, $(6,3,3,2)$	97508421, 63317664

4 The Kaprekar gap simplex for sorted logit spectra

The preceding section concerns sorted *digit* gaps. We now transfer the same viewpoint to sorted *logit* gaps.

Definition 4.1 (Sorted top-k logit spectrum)**.** Let $z \in \mathbb{R}^V$ be a logit vector. Fix $k \geq 2$, take the top-k logits, and sort them as

$$s_1 \geq s_2 \geq \cdots \geq s_k.$$

Assume $s_1 > s_k$, and define the total spread

$$D := s_1 - s_k.$$

Definition 4.2 (Kaprekar gap simplex coordinates)**.** The *Kaprekar gap simplex* (KGS) coordinates of the sorted top-k logits are

$$u_j := \frac{s_j - s_{j+1}}{D}, \qquad j = 1, \ldots, k-1.$$

These are simply normalized adjacent gaps, but the Kaprekar point of view is that cumulative head–tail contrasts become linear functionals of u.

Theorem 4.3 (Simplex property and reconstruction)**.** *Let $u = (u_1, \ldots, u_{k-1})$ be the KGS coordinates of $(s_1, \ldots, s_k)$. Then:*

(i) $u_j \geq 0$ for all j and

$$\sum_{j=1}^{k-1} u_j = 1.$$

Hence u lies in the standard simplex

$$\Delta_{k-1} = \left\{ u \in \mathbb{R}_{\geq 0}^{k-1} : \sum_{j=1}^{k-1} u_j = 1 \right\}.$$

(ii) The sorted top-k logits are reconstructed from (D, u) up to an additive constant via

$$s_i = s_k + D \sum_{j=i}^{k-1} u_j, \qquad i = 1, \ldots, k.$$

Proof. Since $s_j \geq s_{j+1}$ and $D > 0$, each $u_j \geq 0$. Moreover,

$$\sum_{j=1}^{k-1} u_j = \frac{1}{D} \sum_{j=1}^{k-1} (s_j - s_{j+1}) = \frac{s_1 - s_k}{D} = 1.$$

For the reconstruction formula,

$$s_i - s_k = (s_i - s_{i+1}) + (s_{i+1} - s_{i+2}) + \cdots + (s_{k-1} - s_k) = D \sum_{j=i}^{k-1} u_j.$$

□

Corollary 4.4 (Completeness on the affine quotient)**.** *Let*

$$\mathcal{Q}_k := \{ s \in \mathbb{R}^k : s_1 \geq \cdots \geq s_k, \ s_1 > s_k \} / \sim$$

where $s \sim t$ iff $t = as + b\mathbf{1}$ for some $a > 0$ and $b \in \mathbb{R}$. Then

$$[s] \mapsto u(s)$$

is a bijection from $\mathcal{Q}_k$ onto Δ_{k-1}.

Proof. The KGS coordinates are invariant under $s \mapsto as + b\mathbf{1}$ with $a > 0$, so the map is well-defined on the quotient. By Theorem 4.3, every class has a unique representative with $s_k = 0$ and $s_1 = 1$, namely

$$s_i = \sum_{j=i}^{k-1} u_j.$$

This gives surjectivity and uniqueness simultaneously. □

Definition 4.5 (Kaprekar cumulative profiles)**.** For $2 \le r \le \lfloor k/2 \rfloor$, define

$$H_r := \frac{s_r - s_{k+1-r}}{D}.$$

(We exclude $r = 1$ because $H_1 = 1$ is tautological.)

Proposition 4.6 (Cumulative head–tail representation)**.** *For every* $2 \le r \le \lfloor k/2 \rfloor$*,*

$$H_r = \sum_{j=r}^{k-r} u_j.$$

Proof. By telescoping,

$$s_r - s_{k+1-r} = (s_r - s_{r+1}) + \cdots + (s_{k-r} - s_{k+1-r}) = D \sum_{j=r}^{k-r} u_j.$$

Divide by D. □

Remark 4.7 (What is genuinely informative?)**.** The complete affine-invariant shape of the sorted top-k logits is already contained in u. The spread D restores scale, and the cumulative profiles H_r are *redundant but interpretable* coarse summaries. This corrects a subtle but important point: the fully normalized outermost head–tail gap is always 1, so one should not mistake it for a useful feature.

5 Consequences for LLM reliability and decoding

5.1 Why logits alone cannot be universally sufficient

Current reliability papers increasingly combine output statistics with internal-state probes or conformal wrappers [29, 27, 31, 30]. The next proposition explains why some such enrichment is mathematically unavoidable.

Proposition 5.1 (No universal logit-only correctness score)**.** *Fix any measurable score* $F : \mathbb{R}^V \to \mathbb{R}$ *that depends only on a current logit vector* z*. Then there is no theorem of the form "*$F(z)$ *is a universally sufficient statistic for correctness" valid across all tasks or data-generating processes.*

Proof. Choose any $z_0 \in \mathbb{R}^V$. Define two data-generating processes on pairs $(Z, Y) \in \mathbb{R}^V \times \{0, 1\}$ by

$$\mathbb{P}_P(Z = z_0, Y = 1) = 1, \qquad \mathbb{P}_Q(Z = z_0, Y = 0) = 1.$$

The logit marginal is identical under P and Q, so $F(Z)$ is identical as well, yet the conditional correctness probability differs maximally. Therefore no function of logits alone can be universally sufficient for correctness without additional structural assumptions. □

Proposition 5.1 does *not* imply that logit geometry is useless. It implies that a practical system should either (a) combine KGS with internal states, or (b) calibrate its decisions against held-out labeled data, or both.

5.2 A benchmark-ready hybrid architecture

For generation step t, let D_t and u_t be the spread and KGS coordinates of the sorted top-k logits, and let H_t be a short vector of cumulative profiles $H_{t,2}, \ldots, H_{t,R}$. Define

$$\phi_t = \big[\log(1 + D_t),\ u_{t,1}, \ldots, u_{t,J},\ H_{t,2}, \ldots, H_{t,R}\big].$$

A natural token-level uncertainty head is

$$g_t = \sigma(w^\top \phi_t + b),$$

where σ is the logistic function.

To make the score competitive with the current state of the art, Proposition 5.1 suggests selected-token hybridization with hidden states. Let h_t be a frozen hidden representation from the LLM at step t, and define attention weights

$$a_t = \frac{\exp(\lambda\, v^\top \phi_t)}{\sum_{s=1}^{T} \exp(\lambda\, v^\top \phi_s)}.$$

Then aggregate

$$h^\star = \sum_{t=1}^{T} a_t h_t$$

and output sequence-level confidence

$$C(x) = \sigma\big(w_c^\top [h^\star, \mathrm{pool}(\phi_1, \ldots, \phi_T)] + b_c\big).$$

This architecture is deliberately aligned with the current frontier: cheap logit geometry, selected-token internal probes, and external calibration [27, 28, 30, 31, 29].

5.3 A Kaprekar-guided relaxed verification rule

Let $s_1 \geq \cdots \geq s_k$ be target-model top-k logits at a verification step, let $D = s_1 - s_k$, and suppose a draft token has target rank $r \leq k$. Define the *Kaprekar margin acceptance rule*

$$\frac{s_1 - s_r}{D} \leq \rho \qquad (0 \leq \rho \leq 1).$$

This accepts tokens whose normalized logit loss relative to the top token is at most ρ.

Proposition 5.2 (Probability-ratio guarantee)**.** *Under the rule above,*

$$\frac{p_r}{p_1} \geq e^{-\rho D},$$

where p_i is the target softmax probability of the token with rank i. Consequently,

$$p_1 - p_r \leq p_1\big(1 - e^{-\rho D}\big).$$

Proof. The acceptance condition is equivalent to

$$s_r - s_1 \geq -\rho D.$$

Exponentiating yields

$$e^{s_r - s_1} \geq e^{-\rho D}.$$

Since softmax ratios satisfy $p_r/p_1 = e^{s_r - s_1}$, the first claim follows. The second is then

$$p_1 - p_r = p_1 \left(1 - \frac{p_r}{p_1}\right) \le p_1(1 - e^{-\rho D}).$$

$\square$

Proposition 5.2 is not a semantic-quality guarantee. It is a mathematically transparent *distributional* guarantee: accepted alternatives remain within a controlled normalized logit band. This is the right level of theorem to attach to speculative decoding without pretending to have proved end-task preservation.

5.4 Conformal wrapping

Once a sequence-level score $C(x)$ is obtained, it can be wrapped by a selective conformal procedure such as SConU or newer adaptive variants [24, 37]. The finite-sample guarantees then come from those external calibration frameworks, not from KGS itself. This division of labor is important: KGS supplies a structured representation, whereas conformal machinery supplies the formal coverage or risk guarantees.

6 Computational verification

Theorems 3.4 and 3.7 are exact and require no computation for validity. Nevertheless, exhaustive verification is useful for catching implementation mistakes and for documenting the reduced-state picture.

We exhaustively enumerated every digit string for all pairs

$$2 \le b \le 8, \qquad 1 \le n \le 7,$$

a total of 49 finite cases, and in each case verified:

1. the factorization identity $K_{b,n}(x) = \Lambda_{b,n}(\Sigma(x))$ for every state x;
2. the image-cardinality formula
$$|\operatorname{Im}(K_{b,n})| = \binom{b + \lfloor n/2 \rfloor - 1}{\lfloor n/2 \rfloor};$$
3. the reduced-map recovery of the decimal fixed spectra in Table 1.

Representative image-cardinality checks appear in Table 2. The verification script is included with this manuscript as supplementary code.

7 Relation to the 2026 state of the art and a benchmark protocol

This paper is theorem-first, so it does *not* claim state-of-the-art empirical performance. However, it is designed to be tested directly against the current frontier. The strongest current trend is not "entropy only" and not "logits only"; it is the combination of cheap output-side signals, selective internal probes, and deployment-facing calibration [29, 27, 30, 31, 24, 37]. KGS is intended to be the output-side geometric component in exactly that hybrid stack.

A rigorous benchmark suite should compare at least the following families:

Table 2: Representative exhaustive checks of Theorem 3.7.

Base b	Digits n	Predicted $\lvert\mathrm{Im}(K_{b,n})\rvert$	Exhaustively observed
2	4	3	3
3	5	6	6
5	6	35	35
8	7	120	120

1. **Output-only baselines:** max softmax probability, entropy, top-k entropy, top-1 margin, and LogU [22].

2. **Meaning-space baselines:** semantic entropy and semantic-entropy probes [19, 20].

3. **Internal-state baselines:** hidden-state probes, attention-head entropy, local-information scores, and one-token verification where applicable [18, 30, 31, 32].

4. **Hybrid/selective baselines:** entropy-plus-probe models and conformally calibrated abstention systems [29, 24, 37].

The benchmark tasks should include open-domain factual QA, multi-hop QA, medical QA, and reasoning benchmarks. The evaluation should report at least AUROC, AUPRC, Brier score, expected calibration error, and risk–coverage curves. For speculative decoding, the relevant metrics are average acceptance length, realized tokens per second, and downstream-quality change relative to exact verification.

The benchmarkable predictions of this paper are:

P1: KGS should outperform scalar entropy and scalar margin in regimes where the *shape* of the top-k head matters, not just its total spread.

P2: Hybrid KGS-plus-probe systems should outperform KGS alone under distribution shift, in accordance with Proposition 5.1.

P3: The Kaprekar margin acceptance rule should improve acceptance length in flatter-head regimes relative to top-2-only margin rules, while preserving a transparent distributional guarantee via Proposition 5.2.

Any claim stronger than **P1**–**P3** requires an actual multi-model benchmark and should not be asserted without it.

8 Limitations and scope

The manuscript is intentionally conservative in four ways.

First, the factorization theorem is elementary, and some pieces of the underlying idea are likely implicit in earlier difference-coordinate literature [8, 13, 14]. The novelty claim should therefore be read as follows: we are isolating the quotient structure, proving the exact image-cardinality theorem, and transferring the geometry to LLM logits in a mathematically clean way.

Second, the LLM sections are *benchmark-ready* rather than benchmark-complete. They define a representation, prove structural properties, and specify falsifiable tests; they do not report a multi-model empirical study.

Third, KGS is a representation, not a universal oracle. Proposition 5.1 shows why this limitation is fundamental.

Fourth, conformal guarantees require calibration assumptions such as exchangeability or justified shift-handling procedures. KGS does not remove that requirement; it only provides a structured score to calibrate.

9 Conclusion

Kaprekar's constant 6174 is not the deepest object in the story. The deeper object is the ordered-gap law behind the routine. In the digit setting, that law yields an exact reduced-state representation and a closed formula for the one-step image size. In the logit setting, the same law produces a simplex geometry that cleanly separates spread from shape and exposes which aspects of a top-k spectrum are affine-invariant.

The central message of the paper is therefore concise:

Do not replace 6174 by a new magic number. Replace it by a gap-spectral equation.

That equation is mathematically precise, computationally useful, and directly relevant to the current frontier in LLM reliability, routing, and decoding.

A Chronological literature map (selected, 1949–2026)

Year	Contribution	Reference
1949	Original digit-rearrangement solitaire game.	[1]
1955	Classical 6174 note.	[2]
1964	Early self-producing digit-sequence analysis.	[3]
1978	Classification of all four-digit Kaprekar constants.	[4]
1981	Proof that the only nonzero decimal constants are 495 and 6174.	[5]
1993	Two-digit variant and loop analysis.	[6]
2004	Widely cited pedagogical synthesis.	[7]
2005	Search across bases and higher digit lengths.	[8]
2017	Base-dependent regularities and symmetry viewpoint.	[9, 10]
2018	2-adic constants and two-digit distances.	[11]
2019	Explicit formulas and higher-digit generalization in arbitrary base.	[12, 13]
2022	Symmetry/parametric work in decimal and a broad odd-base theory.	[14, 15]
2024	Broad even-base classification.	[16]
2026	Entropy, drift fields, and attractor-basin analysis.	[17]
2023	Hidden-state truthfulness probing for LLMs.	[18]
2024	Semantic entropy and semantic-entropy probes.	[19, 20]

Year	Contribution	Reference
2025	Large-scale UQ benchmarking; logit UQ; query-level uncertainty; judge decoding; reasoning-model calibration.	[21, 22, 23, 25, 26, 27, 28]
2026	Hybrid selective prediction, attention-entropy predictors, local-information scores, one-token verification, logit-monitoring, direct acceptance optimization, and adaptive conformal calibration.	[29, 30, 31, 32, 33, 34, 35, 37]

B Supplementary reproducibility note

The supplementary Python script included with this manuscript verifies:

1. Theorem 3.4 for all $2 \leq b \leq 8$ and $1 \leq n \leq 7$ by exhaustive enumeration.
2. Theorem 3.7 on the same grid.
3. The decimal fixed spectra displayed in Table 1.

The script is intentionally compact and uses only elementary arithmetic; it is meant as a correctness audit rather than as a high-performance implementation.

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